

Clockster

Attendance & Discipline Analysis

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Methodology

Steps that were taken in analyzing the data :

1. Identify the problems.
2. Set data analysis goals.
3. Data gathering & collection
3. Data cleaning and manipulation.
4. Analysis and visualization.
6. Insights and recommendation

Project Description

The HR department provides employee attendance data including schedules, logs, leave requests, and employee details.

Objectives

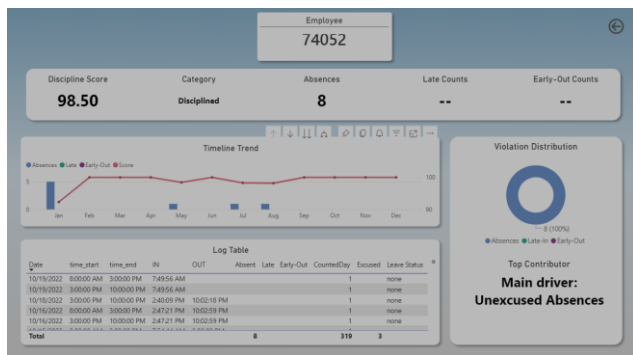
1. Identify the most and least disciplined employees and department.
2. Analyze weekly and monthly behavior pattern.
3. Evaluate departmental tolerance toward undisciplined behavior.

Most & Least Disciplined Employees

The top 3 consistently reflect disciplined behavior.

Top 3 Discipline

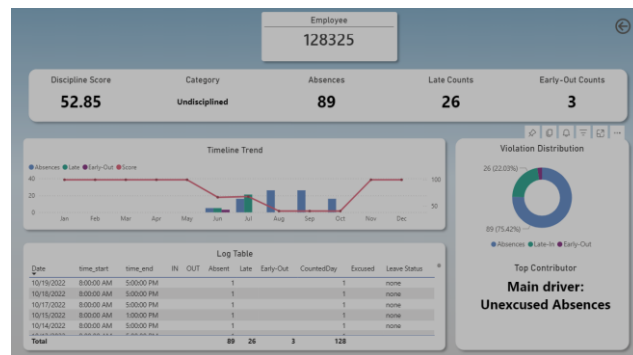
Emp ID	Rank	Score
74052	★ 1	98.50
74461	☆ 2	97.68
83884	☆ 3	97.66



Bottom 3 employee demonstrate the highest violation rates, particularly absences as the main driver.

Bottom 3

Emp ID	Rank	Score
128325	🚩 30	52.85
75218	🚩 29	56.57
88347	🚩 28	58.04



Scores are exposure-based; accepted/pending leave excluded

Most & Least Disciplined Department

Pharmacy ranks as the most disciplined department, while PBF shows the lowest discipline score, indicating concentrated attendance issues in that group.

The ranking here is raw (includes all employees in each department), which helps surface hotspots even if a department contains low-exposure or extreme-absence cases.

Department Discipline

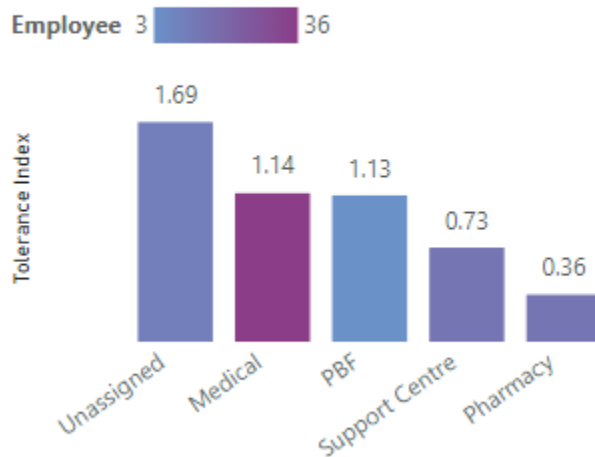
Department	Rank	Score
Pharmacy	★ 1	90.19
Support Centre	★ 2	81.90
Medical	🚩 3	72.49
PBF	🚩 4	54.38
Unassigned	🚩 5	41.89

Which departments appear more forgiving?

Unassigned shows the highest tolerance index, signaling a higher share of undisciplined employees than the company baseline.

- Index > 1.0 means department's undisciplined share is above company share.
- Index < 1.0 means lower than the company baseline.

Tolerance Index by department



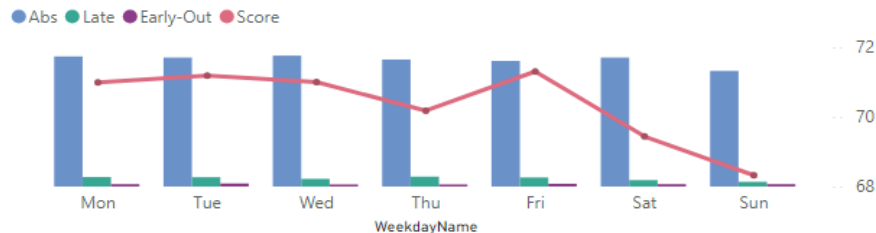
Weekly and Monthly Behavior Patterns

Violations peak on Mondays (absences) and Thursdays (lateness); month-level spikes occur in March and May, while Nov. & Dec. show weaker discipline.

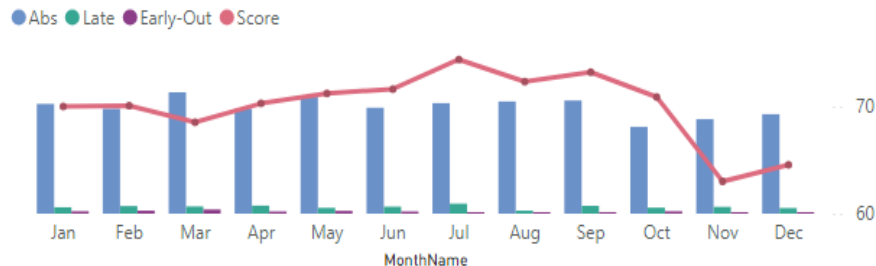
Weekly: Monday = absences;
Thursdays = late

Monthly: March & May = absences;
Nov. & Dec. = low score

Violation & Score Trend



Violation & Score Trend



Counts and rates are normalized by shift-based; accepted/pending leave is excluded.

Summary of Findings

- **Employees:** Reliable ranking identifies Top 3 and Bottom 3 by 0—100 score.
- **Departments:** Pharmacy most disciplined; PBF lowest score (raw ranking).
- **Timing:** Monday = most absences; Thursday = most lateness;
March/May spikes; Nov—Dec lowest scores.
- **Tolerance:** Unassigned has the highest tolerance index (> 1.0).

Summary of Insights

- Exposure-aware scoring prevents unfair comparisons; absence drives score most.
- Reliability gates (min exposure; absence cap) stabilize rankings for action.
- Hotspots emerge in specific departments and calendar windows.
- Tolerance Index distinguishes prevalence issues (culture/enforcement) from single outliers.

Recommendations

- **Targeted governance:** Prioritize Unassigned and PBF for attendance policy reinforcement and tagging review.
- **Time-based actions:** Monday attendance reminders; Thursday punctuality checks; staffing/coverage for March/May.
- **Workflow & alerts:** Automate leave approvals, violation alerts, and manager follow-ups.
- **Maintain fairness:** Keep reliability gates in reporting to guide coaching, not penalize noise.